

SIP-7701

Native Account Abstraction

Why Account Abstraction?

Account Abstraction

- Smart Contracts can define validity of transaction payload - no EOAs involved
 - Passkeys
 - Session keys
 - Multisig signatures
 - Post-Quantum signatures
- Enable “gas abstraction”
 - Gas fees sponsored by 3rd party
 - Users charged with SRC-20
 - Privacy protocols withdrawal
- Execution is defined by the Smart Contracts’ implementation

SRC-4337 Market Fit

SRC-4337 Metrics

Total UserOps

238,994,323

UserOps

Total Bundle Txns

142,813,625

Transactions

Total Paymaster Volume

\$6,702,668

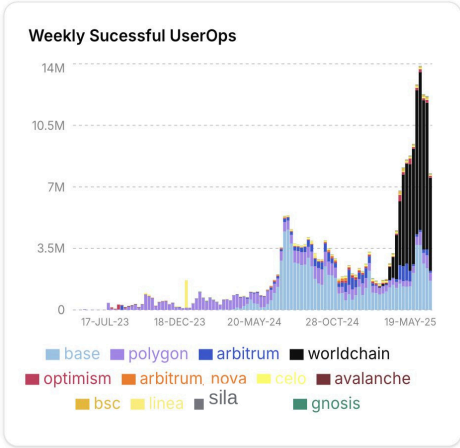
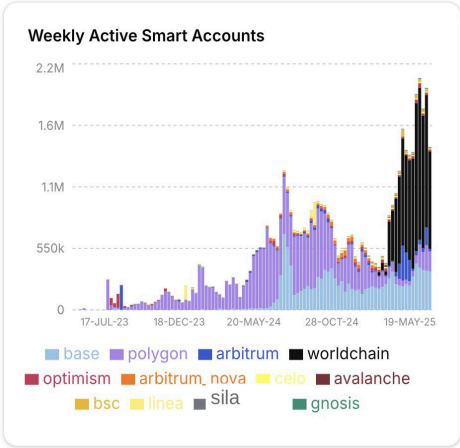
Gas Covered

Accounts with 1+ UserOps

32,531,327

Accounts

WEEKLY CHARTS



Know Nothing Labs	Cometh	OKX
Fun Wallet	AnonAA	Schnorrkel.js
MynaWallet	IoTeX	Ambire Wallet
MetaMask	Ledger	Coinbase
Alchemy	Biconomy	Argent
WalletConnect	Candide Wallet	Trampoline
BananaHQ	zkShield	ZeroDev
Soul Wallet	ThirdWeb	UniPass Wallet
Safe	Lemon Wallet	Obvious Wallet
Toyota Blockchain Lab+	Privy	Trust Wallet
Silaspot	Clave	Circle
Frame Wallet	Rainbow	Rabby
Zerion	Status Wallet	+ New page

SRC-4337 Lessons Learned

- Enthusiastic adoption by wallets and dapps
 - Safe, OKX, Trust, Ambire, Coinbase, Soul, ZeroDev etc.
- Gas Sponsoring and Abstraction is a major case
 - 85% of UserOperations used a Paymaster
 - Sponsored operations are big source of adoption
 - Paymasters simplify new user onboarding
- Gas overhead hinders adoption
 - Using a private (centralized) relay is cheaper
 - Aggregated overhead is costly even on L2s

Paymaster	Total UserOps Served	↑↓	Total Gas Spend	↑↓
pimlico	40,638,145		\$1,691,982.04	
Unknown	24,038,969		\$1,252,914.89	
biconomy	17,277,178		\$1,164,222.83	
alchemy	119,509,841		\$1,029,798.35	
stackup	536,472		\$815,192.31	
coinbase	27,441,116		\$748,894.62	
particle	467,586		\$77,622.27	
circle	2,134,042		\$15,709.37	
send	909,379		\$10,120.99	
thirdweb	1,409,760		\$8,223.88	
candide	32,554		\$2,344.19	
ambire	98,539		\$1,139.73	
blocto	1,315		\$1,018.45	
cometh	108,176		\$297.76	
nani	482		\$16.60	
silaspot	378		\$12.94	

Why Native AA?

Native AA: Requirements

- Smart Contracts can define validity of transaction payload - no EOAs involved
- Enable “gas abstraction”: SIL paid by 3rd party, users charged SRC-20, privacy
- New Smart Contract can be deployed using Native AA without EOAs
- SIP-7870 hardware requirements are preserved for all nodes
- Avoid unnecessary complexity
- Avoid regressions

Native AA: Advantages

- Removes Gas Overhead
 - Gas overhead makes centralized relays more attractive
- Enables Censorship Resistance
 - FOCIL can support Native AA transactions
 - No need for centralized (censoring) “relay” or “bundler”
- Enables Post Quantum accounts
 - No need for PQ-EOA transaction type
 - No “migration” transaction - can use SIP-7702

Native AA: Gas Savings

SRC-4337:

- Static overhead: ~30'000 gas
- Per-operation overhead: ~40'000

This overhead appears to be inherent to the non-native approach.

SIP-7701 total overhead estimation: ~5'000

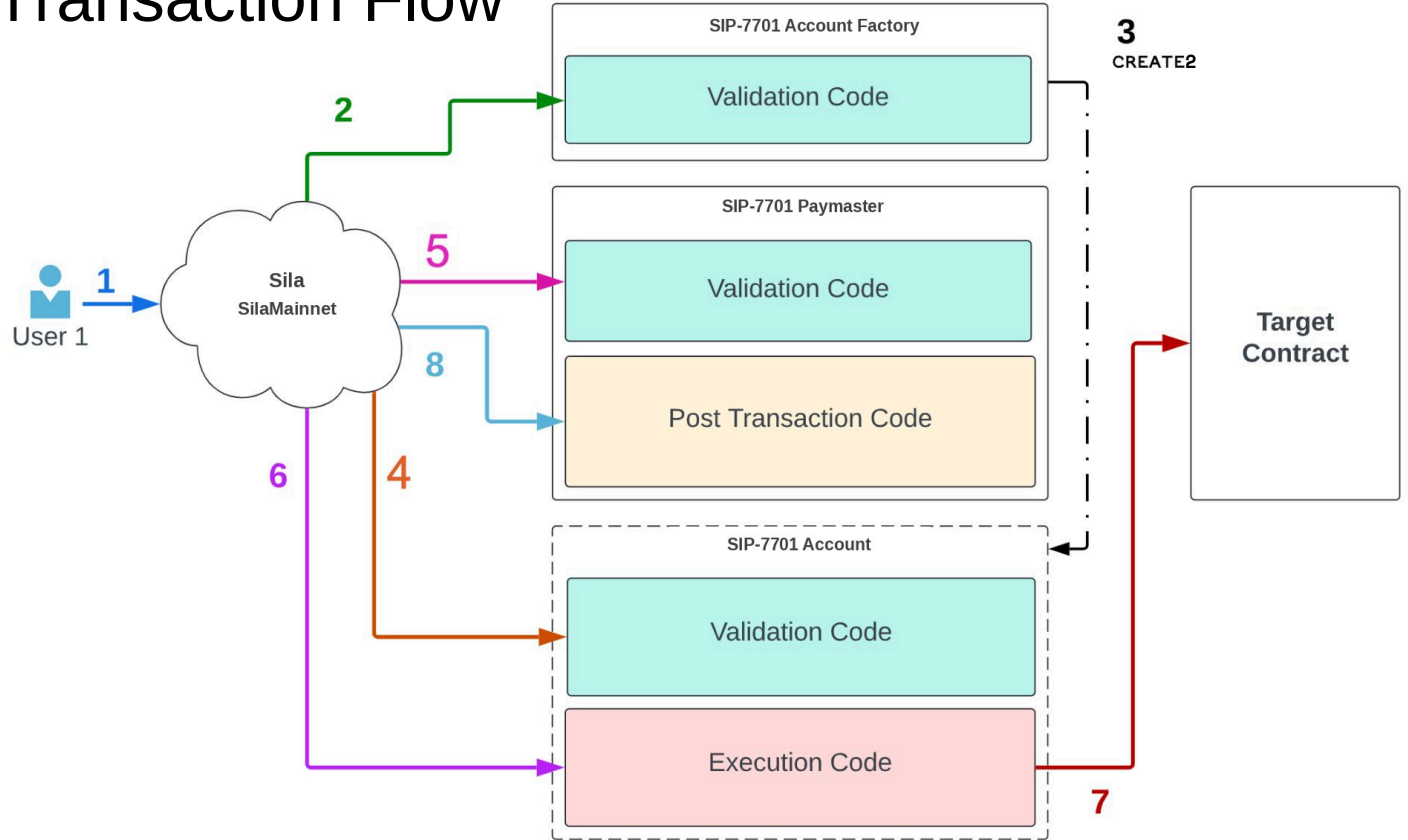
*measured for simple SRC-20 transfer - EOA vs ECDSA Smart Account

Why SIP-7701?

SIP-7701 Smart Contract Roles

- Sender Account
 - Validates the transaction
 - Pays the gas (if there is no Paymaster)
 - Executes the actual calls
- Deployer
 - Creates the Account (if necessary)
- Paymaster
 - Pays the gas (if specified)

SIP-7701 Transaction Flow



SIP-7701 Smart Contract Roles - why?

- Separating “validation” from “execution” required to prevent DoS on nodes
 - Limiting “validation” gas - a “potentially unpaid” work for nodes
 - Enable sandboxing the “validation” code to prevent N invalidations at $O(1)$ cost
- Composability - Account and Paymaster are separate entities
 - Enables “mutual distrust”
 - Prevents potential Paymaster-vs-Account attack vectors

SIP-7701 Transaction Type

```
AA_TX_TYPE || rlp([
    chain_id,  nonce,
    sender, sender_validation_data,
    deployer, deployer_data,
    paymaster, paymaster_data,
    sender_execution_data,
    max_priority_fee_per_gas, max_fee_per_gas,
    sender_validation_gas, paymaster_validation_gas,
    sender_execution_gas, paymaster_post_op_gas,
    access_list,  authorization_list
])
```

SIP-7701 Opcodes

- **TXPARAMSIZE, TXPARAMCOPY, TXPARAMLOAD**

- Available in the top-level frame
- Provide access to all parameters of the **AA_TX_TYPE** transaction

- **CURRENT_ROLE**

- Provides access to the identifier of the current role

- **ACCEPT_ROLE**

- Indicates the calling contract accepts proposed role in the current **AA_TX_TYPE** transaction

SIP-7701 Transaction Validity

Protocol:

- ✓ Sufficient gas balance (account or paymaster)
- ✓ Correct account nonce

EVM Code Execution:

- ✓ Deployer accepted its role (first transaction for account)
- ✓ Account is deployed
- ✓ Account accepted its role
- ✓ Paymaster accepted its role (if paymaster specified)

SIP-7701 Transaction Mempool

- DoS possible if validation code has arbitrary state access
 - Mass Invalidation - i.e. one SSTORE invalidates multiple transactions
 - Becomes a big DoS vector
- Validation code limitations not part of core SIP - node configuration
- **SRC-7562** (not final) - set of limitations for validation code
 - Contracts only access “their own” storage
 - Ban environment introspection (time, blocks, gas, balances etc.)
 - Gas Limits for validation frames
 - Requires nodes to trace incoming SIP-7701 transactions’ validation frames

SIP-7701 with FOCIL and Statelessness

- FOCIL allows excluding IL transactions if invalid
- Checking validity of Native AA transaction consumes up to `MAX_VALIDATION_GAS`
- Checking validity of Native AA transaction requires bytecode and state
- Mitigations:
 - Reasonable limits on maximum total IL validation gas in AA-FOCIL
 - Reasonable limits on maximum witness sizes in AA-VOPS
 - Include contracts bytecodes in “active” storage for AA-VOPS

See the “Stateless” session on Friday!

SIP-7701: Alternatives

- Do nothing
 - AA remains SRC-4337 only
 - SRC-4337 UserOperations get no CR through FOCIL
 - Self-relaying possible but violates privacy
 - High gas overhead hinders adoption
 - For now, no solution for Post-Quantum accounts
- Make FOCIL aware of SRC-4337 UserOperations
 - Mixes application layer and protocol layer
 - All other issues mentioned above
- Another Native AA proposal supporting fewer use-cases

Extras

SIP-7701 Block building

- Block building and re-validation computation overhead
 - Each new cold SSTORE can invalidate only ONE transaction in mempool (SRC-7562)
 - Each mempool transaction can take up to **MAX_VALIDATION_GAS** to re-check
 - E.g. a transaction of 5'000'000 gas may cause **1'000 x MAX_VALIDATION_GAS gas** re-check
- Mitigations
 - Must select a **MAX_VALIDATION_GAS** very cautiously
 - Consider additionally limiting the number of SLOADs in the validation code
 - Consider using SIP-2930 'access lists' enforcement in validation code
 - allows builders to delay re-checking affected transactions